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With increasing competition in the global marketplace, international segmentation has become an ever more important issue in developing, positioning, and selling products across national borders. The authors propose a methodology to identify cross-national market segments, based on means–end chain theory. The methodology offers the potential for integrating product development and communication strategies by linking product characteristics to consumer benefits and values. For that purpose, a model is developed that identifies relations between the consumer and the product at the segment level, which increases the actionability and responsiveness of the segments. The model accounts for different response tendencies, across and within countries, that commonly hamper identification of cross-national segments. A Monte Carlo study shows that the model performs well in recovering the parameters across a wide range of conditions. The segmentation model is applied to consumer data on yogurt collected in 11 countries of the European Union. Four international segments are identified and found to be related to consumer sociodemographics, consumption patterns, media consumption, and personality. The authors show that the model has high predictive validity and outperforms the standard clustering approaches traditionally employed in international segmentation.

International Market Segmentation Based on Consumer–Product Relations

The globalization of the marketplace is arguably the most important challenge facing companies today (Yip 1995). Developments accelerating the trend toward global market unity include rapidly falling national boundaries; regional unification (e.g., the European Union [EU], North American Free Trade Agreement, Association of Southeast Asian Nations, Mercosur); standardization of manufacturing techniques; global investment and production strategies; expansion of world travel; rapid increase in education and literacy levels; growing urbanization among developing countries; free flow of information (e.g., World Wide Web), labor, money, and technology across borders; increased consumer sophistication and purchasing power; advances in telecommunication technologies; and the emergence of global media (Alden, Steenkamp, and Batra 1999; Hassan and Katsanis 1994; Mahajan and Muller 1994). Firms cannot ignore this trend toward globalization. Even if they decide not to be involved in the global (or pan-regional) marketplace, companies still face increased competition in their home markets because of nimble foreign competitors reaping the benefits of global strategies (Yip 1995). The marketing environment has become competitive to an extent that requires firms to target their products at markets that span national boundaries. However, competitive clout cannot be achieved in global consumer markets unless firms thoroughly understand and adequately respond to the core values and needs of those consumers (Hassan and Kaynak 1994).

Globalization affects consumer behavior and attitudes in many ways, in that they transcend national borders. Groups of consumers in different countries often have more in common with one another than with other consumers in the same country. Therefore, there is a greater receptivity to global brands and foreign products across the world. Consumer brands have received wide global acceptance in categories such as consumer electronics, cars, fashion, home ap-

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pliances, food products, and beverages (Hassan and Katsanis 1994). Many of these products respond to the needs and wants of segments of consumers that cut across national boundaries (Hassan and Kaynak 1994).

Thus, a major challenge facing international marketers is to identify global market segments and reach them with products and marketing programs that meet the common needs of these consumers (Hassan and Katsanis 1994). This requires that characteristics of the (physical) product configuration are developed and marketed with the specific desires of the global target segment as guiding principles. Such a consumer-oriented approach is critical to the success of global products and brands (Cooper 1984). Kleinschmidt and Cooper (1988) find that products designed for the global market achieve almost twice the market shares of products with domestic design that are aimed at the same overseas markets. In a similar vein, Wind and Mahajan (1997) emphasize the importance of a global scope in product strategy. Research should be conducted in multiple countries rather than just the home country (cf. Steenkamp, ter Hofstede, and Wedel 1999). This requires new developments in cross-national research methodology and models that integrate multicountry data at a pan-regional or global segment level (Wind and Mahajan 1997). It is in this area that this study is situated.

The purpose of this article is to propose an integrated methodology to identify segments in international markets based on consumer means–end chains (MECs). The key idea underlying MECs is that product attributes are means for consumers to obtain desired ends, namely, values, through the benefits yielded by those attributes (Gutman 1982; Newell and Simon 1972; Reynolds and Olson 1998). In MEC theory, these three concepts are linked hierarchically in cognitive structures, in that product attributes yield particular benefits upon consumption, which contribute to value satisfaction. Means–end chain theory provides a conceptual basis for linking the product and consumer in an international context, which facilitates successful product development and communication strategies (Gutman 1982). The interrelations among attributes, benefits, and values can differ across consumers within countries and also across countries.

The methodology for identifying international consumer segments integrates data collection with a model for data analysis. The model is tailored to binary MEC data. Several features that are important for identifying valid international market segments are accommodated in our model, as follows:

1. The model captures the means–end relations that form the basis of cross-national segments in a stochastic framework. In the model, probabilistic relations between attributes, benefits, and values are specified at the segment level.
2. The model captures differences in response behavior that may exist among consumers in different countries, which is an important issue in comparing and grouping data across countries (Chen, Lee, and Stevenson 1995; Steenkamp and Baumgartner 1998). The model also captures within-country heterogeneity in response behavior that may exist among consumers.
3. In modeling the segment sizes, we use a specification in which country memberships enter as concomitant variables. This enables the segment sizes to differ across countries, so that truly pan-regional or global segments are accommodated.
4. We use a pseudo maximum likelihood approach to arrive at unbiased estimates of the model parameters. Because international marketing research generally deals with samples stratified by nation, with equal allocation of sample sizes, traditional estimation models result in biased parameter estimates.

Although the model is specifically suitable for international market segmentation, it nests situations of domestic segmentation, that is, segmentation in a single country. In the "Discussion" section, we show how domestic segmentation is accommodated as a special case of our model.

To assess the performance of the model, we include a Monte Carlo analysis of synthetic data. We apply the model to data from a large-scale international consumer survey sponsored by the European Commission. The sample comprises nearly 3000 consumers from 11 countries of the EU. We discuss the results and their implications for marketing strategy. We then relate the segments to descriptor variables and assess the predictive validity. We evaluate the incremental contribution of the model over traditional approaches both conceptually and empirically. Finally, we present conclusions, limitations, and issues for further research.

THE INTERNATIONAL SEGMENTATION BASIS

In international segmentation studies, information at the country level often has been used as a basis for grouping countries. Macro-level geographic, political, economic, and cultural data typically have been used to identify market segments that consist of groupings of countries (e.g., Helsen, Jedidi, and DeSarbo 1993; Kale 1995). This approach provides insights into which groups of countries potentially can be targeted, but it provides no information on which consumers in those countries will respond to marketing efforts. Within-country heterogeneity is ignored, in spite of being the very basis for identification of domestic segments. If consumers, instead of countries, were used as the basis for identifying international segments, the effectiveness of marketing strategies would increase greatly (Jain 1989).

Recently, at least two broad classes of micro-level segmentation bases have been used in international segmentation research. Some studies have used product-specific bases, including perceived product characteristics (e.g., Moskovitz and Rabino 1994). Other studies have used consumer-specific bases such as lifestyles (e.g., Boote 1983) and values (e.g., Kamakura et al. 1993). Whereas consumer-specific bases are closer to the consumer, product-specific bases are more actionable (Urban and Hauser 1993). Building on these studies, we propose to incorporate both consumer and product characteristics in international market segmentation. Using measures that connect the consumer and the product creates the potential to increase the effectiveness of global or pan-regional marketing strategies. Such an approach helps link product development to communication strategies that attempt to position products by associating the product aspects to the achievement of desired ends (Gutman 1982).

Newell and Simon (1972) developed the concept of the MEC. Means–end chain theory explicitly establishes the relation between product and consumer by positing hierarchical links between product attributes, the beneficial aspects of product use, and consumers' values: Attributes lead to

benefits, which contribute to value satisfaction. In MEC theory, the dominant meaning of a product attribute to a consumer is determined by the benefits to which it is perceived to lead. Benefits accrue to a consumer when the product is purchased and used. They derive valence and importance from their perceived ability to satisfy personal values. Values underlie a large and important part of human cognition and are among the most central determinants of consumer behavior (Steenkamp, ter Hofstede, and Wedel 1999). They transcend specific objects, in contrast to attributes and benefits, which relate to a particular product (Rokeach 1973). In summary, MEC theory posits that the way in which physical product attributes are linked to personal values defines how products gain personal relevance and meaning. An attribute is important if it leads to a desired benefit, whereas the perceived benefit derives its importance through the extent to which it is linked to one or more personal values (Reynolds and Olson 1998).

Linking products and consumers in MECs for the purpose of international segmentation combines the strengths of product- and consumer-specific bases of segmentation. Identifying segments from concrete product attributes increases the actionability of the results for product development (Urban and Hauser 1993). However, it is well known that consumers buy products for what the products can do for them and how they contribute to achieving desired ends, not for their physical attributes per se. Thus, effective marketing strategies require insights into the links between product attributes, benefits, and values. The benefits associated with product attributes support persuasive claims of the product. Related values help substantiate and frame these claims in targeted advertising campaigns (Reynolds, Gengler, and Howard 1995). Therefore, it is not surprising that several researchers have suggested that MECs provide a particularly suitable basis for market segmentation (Gutman 1982; Kamakura and Novak 1992).

THE INTERNATIONAL SEGMENTATION METHODOLOGY

In our approach, the perceived means–end relations between attributes, benefits, and values are measured at the individual level using a sequence of pick any/J tasks. The underlying idea is to decompose a MEC relation into two independent parts, attribute–benefit (AB) and benefit–value (BV) links. The data collection procedure is called the "Association Pattern Technique" (APT; ter Hofstede et al. 1998). See Appendix A for details on the APT task.

In modeling the data collected in an APT task, we consider the consumer's response process in a probabilistic framework. This is consistent with Gutman's (1991, p. 144) ideas, which state that links "can be thought of in terms of some probability statement that one element will cause the occurrence of the other element." The choice of a link in either of the APT matrices depends on the intrinsic strength of the link. It also depends on the respondent's response tendency, that is, his or her propensity to indicate a link in general, regardless of the underlying strength (importance) of the AB or BV relation. Measurement instruments are subject to response tendencies, and those tendencies may differ across countries (Chen, Lee, and Stevenson 1995; Steenkamp and Baumgartner 1998). International segmentation studies employing consumer surveys are affected adversely by cross-national differences in response behavior. Variation in response tendencies may mask cross-national segments, and the segments identified could constitute groups of consumers that differ in response tendency rather than in the underlying behavior studied. In this study, we explicitly account for differences in response tendency within and across countries.

In our model, segments are represented by a finite mixture formulation (Titterington, Smith, and Makov 1985). Let u_j^m denote the intrinsic strength of link j in matrix m ($j = 1, \ldots, J_m$) and y_{ij}^m a binary variable with $y_{ij}^m = 1$ if subject i ($i = 1, \ldots, I$) chooses link j in the AB ($m = 1$) or the BV ($m = 2$) matrices, and $y_{ij}^m = 0$ otherwise. Analogous to item response theory, it is assumed that if the strength of link j exceeds subject i's threshold ($u_j^m > \theta_i$), he or she chooses the link. Thus, the smaller the threshold, θ_i, the higher the propensity of the subject is to choose any link, whereby θ_i captures the response tendency of subject i. The probability of subject i choosing link j then can be expressed as a function of the strength u_j^m and the threshold parameter θ_i. We assume the cumulative distribution of u_j^m is logistic, so that we obtain the following closed-form expression for the probability of observing a link j for subject i:

$$(1)\quad P\left(y_{ij}^m = 1 | u_j^m, \theta_i\right) = P\left(u_j^m > \theta_i\right) = \frac{\exp\left(u_j^m - \theta_i\right)}{1 + \exp\left(u_j^m - \theta_i\right)}.$$

For reasons of identification, the threshold parameters are restricted to sum to 0, namely, $\Sigma_i \theta_i = 0$. Equation 1 corresponds to the Rasch model as used in item response theory. Equation 1 shares the following properties with the Rasch model: Both u_j^m and θ_i are defined on the real axis. A higher u_j^m means that the particular link is stronger, and the higher θ_i, the lower the propensity is that a person chooses links in general. In addition, the distance between u_j^m and θ_i may be interpreted in terms of log odds ratios, that is, the logarithm of the ratio of the probability of observing a particular link to the probability of not observing that link.

We use the strength of the link as the basis for segmentation. For that purpose, we let u_j^m vary over S unobserved segments ($s = 1, \ldots, S$). Given segment s, the subject-specific density is the following:

$$(2)\quad f_{i|s}(y_i | \theta_i; u_s) = \frac{\exp\left[\sum_{m=1}^{2}\sum_{j=1}^{J_m} y_{ij}^m (u_{js}^m - \theta_i)\right]}{\prod_{m=1}^{2}\prod_{j=1}^{J_m}\left[1 + \exp(u_{js}^m - \theta_i)\right]},$$

where $u_s = (u_{js}^m)$, and $y_i = (y_{ij}^m)$.

There are, however, several problems associated with this formulation. The fixed effects estimator of the incidental threshold parameter θ_i is inconsistent (Ghosh 1995) and is not identified at the individual level (Lindsay, Clogg, and Grego 1991). Yet, not accounting for the heterogeneity in response tendencies may confound the estimates of the

segment-specific link strengths u_s with response tendencies. A possible solution would be to let θ_i vary across the S unobserved segments; that is, $\theta_i = \theta_s$. However, this approach has several disadvantages. It is not evident a priori that the thresholds obey a segment structure similar to the link parameters u_s. In addition, for many applications, such an approach would be too restrictive, in the sense that it does not deal with within-segment heterogeneity because all subjects within a segment are constrained to have an identical threshold. Research has shown that the assumption of within-segment homogeneity may be overly restrictive and result in a loss of explanatory and predictive power (Allenby, Arora, and Ginter 1998). Most important, the segments identified through such a model would differ in their response tendencies. It is exactly this confounding of response behavior and the structural basis of the segments (i.e., the strengths of the links) that we want to avoid. Therefore, we treat the thresholds as random parameters that vary across the population according to a normal distribution. We employ a specification of the thresholds that separates within- and between-country heterogeneity by assuming that the thresholds θ for country c (c = 1, ..., C) have normal distributions, $\Phi(\theta; \mu_c, v_c)$, with mean μ_c and variance v_c. Whereas Equation 2 presents the conditional distribution of subject responses, given θ_i, the unconditional distributions are obtained by taking the expectation of Equation 2 over the thresholds, as follows:

$$(3)\quad f_{i|s}(y_i; u_s, \mu_{c(i)}, v_{c(i)}) = \int_{-\infty}^{+\infty} f_{i|s}(y_i|\theta; u_s)\, d\Phi\left(\theta; \mu_{c(i)}, v_{c(i)}\right),$$

where c(i) indicates the country membership of subject i. The model simultaneously accounts for between-country differences in response thresholds and within-country threshold heterogeneity.

Equation 3 is conditional on segment s. Letting $\pi_{s|c}$ denote the proportion of consumers in country c belonging to segment s, and using a standard finite mixture formulation, the unconditional distribution (on s) equals:

$$(4)\quad f_i\left(y_i; u, \mu_{c(i)}, v_{c(i)}\right) = \sum_{s=1}^{S} \pi_{s|c(i)} f_{i|s}\left(y_i; u_s, \mu_{c(i)}, v_{c(i)}\right).$$

Because segment sizes may differ among countries, that is, a particular segment is more prevalent in some countries than in others, the segment proportions $\pi_{s|c}$ are allowed to vary across countries through a concomitant variable specification (Wedel and Kamakura 1998, p. 147):

$$(5)\quad \pi_{s|c} = \frac{\exp(\gamma_{sc})}{\sum_{t=1}^{S} \exp(\gamma_{tc})}.$$

By imposing the restrictions $\gamma_{Sc} = 0$ within each country, the proportions sum to 1 across segments.

Estimation and Model Comparison

In international marketing research, samples often are stratified by country, with approximately the same sample size in each country. This results in sample sizes that are not proportional to population sizes. This issue is important when the samples are used for international segmentation purposes. Conventional maximum likelihood estimates may be biased severely when applied to stratified samples. Therefore, in estimating the model, we use a modification of the likelihood approach, pseudo maximum likelihood (PML) (Skinner, Holt, and Smith 1989; Wedel, ter Hofstede, and Steenkamp 1998). The unconditional distribution presented in Equation 4 equals the likelihood contribution of subject i, L_i. The sample log-pseudo likelihood is defined as

$$(6)\quad \ln L_P = \sum_{i=1}^{I} w_i \ln L_i,$$

with the sampling weights w_i equal to the ratio of the population share to the sample share. Maximizing the log-pseudo likelihood in Equation 6 with respect to the unknown parameters γ_{sc}, u_{js}^m, μ_c, and v_c yields consistent estimates of the population parameters (Skinner, Holt, and Smith 1989). We use a quasi-Newton algorithm to maximize the log-pseudo likelihood in Equation 6. Gauss-Hermite integration of order R = 32 is used to evaluate the integral in Equation 3. In that way, Equation 3 reduces to a weighted sum, so that the score functions may be derived in closed form (see Appendix B). A problem related to the numerical maximization of the likelihood is the occurrence of local optima. The solution chosen here is to run the models from a range of randomly chosen starting values for the parameters.

To determine the appropriate number of segments, likelihood ratio (LR) test statistics cannot be used. The asymptotic distribution of the LR statistic is not a chi square because certain regularity conditions do not hold (Aitkin and Rubin 1985). Instead, information criteria have been proposed as heuristics for selecting the appropriate number of segments. These criteria impose a penalty on the log-likelihood that is related to the number of parameters estimated. The classical Akaike's Information Criterion (AIC) is order inconsistent and, therefore, tends to overestimate the correct number of segments. Instead, we use the Bayesian Information Criterion (BIC) and the Consistent Akaike Information Criterion (CAIC) (Bozdogan 1987). The BIC and CAIC are related closely, do not suffer from the drawbacks of AIC, and have better asymptotic properties. They are defined as follows: BIC = $-2\ln L + K\ln I$, and CAIC = $-2\ln L + K[\ln(I) + 1]$, where I denotes the number of observations, and K is the number of free parameters in the model [$K = S(J + C) + C - 1$]. To account for the stratified sample design, lnL is replaced by the log-pseudo likelihood $\ln L_P$ in Equation 6, giving rise to pseudo information criteria, denoted as PCAIC and PBIC (Wedel, ter Hofstede, and Steenkamp 1998). We choose the number of segments for which these modified criteria have a minimum value.

These criteria also are used to test for between- and within-country heterogeneity in response behavior. In testing for the existence of between-country heterogeneity, the PCAIC and PBIC of the (unrestricted) model in Equation 6 are compared with those of a restricted model, where $\mu_1 = \mu_2 = \ldots = \mu_C = 0$. Because the overall mean of the thresholds already has been restricted to 0, the number of free parameters in this restricted model is S(J + C). In testing for the existence of within-country heterogeneity, the unrestricted

model is compared with the model with threshold variance restricted to 0 ($v_1 = v_2 = \ldots = v_C = 0$), which has S(J + C) − 1 free parameters.

Country-Specific Entropy

To assess how well each country fits the international segmentation scheme, we propose to use a normed entropy measure (cf. Ramaswamy et al. 1993). We define the entropy measure for each country separately: $E_{S|c} = 1 + \Sigma_s\Sigma_{i:c(i)=c}\alpha_{is}\ln\alpha_{is}/(I_c \ln S)$, where I_c is the number of subjects in country c, and α_{is} is the posterior probability of segment membership obtained by Bayes' rule, as follows:

$$(7) \qquad \alpha_{is} = \frac{\pi_{s|c(i)} f_{i|s}\left(y_i; u_s, \mu_{c(i)}, v_{c(i)}\right)}{\sum_{t=1}^{S} \pi_{t|c(i)} f_{i|t}\left(y_i; u_t, \mu_{c(i)}, v_{c(i)}\right)}.$$

The entropy for a particular country approximates 1 if the posteriors for all consumers in that country are close to either 0 or 1. If the α_{is} in country c are near 1/S, $E_{S|c}$ tends toward 0. A country with lower entropy has a higher proportion of subjects that are difficult to classify in one of the international segments. In this case, a substantial number of subjects within that country has specific MEC structures that do not agree with those of one of the segments. Therefore, deviations from the pan-regional or global segmentation scheme may be identified. Note that the unconditional entropy measure E_S is biased in the case of stratified samples; that is, large (small) countries will have a too low (high) contribution to the entropy. The country-specific entropy is justified because of its independence from the stratified sample design.

MONTE CARLO ANALYSIS

To assess the performance of the model, we performed a Monte Carlo study in which we applied the model to synthetic data with known parameter values. The design of the Monte Carlo study is based on the following four factors, which are hypothesized to affect parameter recovery:

- PRB: The strength of link probabilities, $p_{js}^m = \exp(u_{js}^m)/[1 + \exp(u_{js}^m)]$, which are generated either in the .3–.7 or the .1–.9 range;
- TSE: The within-country dispersion of thresholds, with standard errors $\sqrt{v_c}$ generated in the range of .3–.5 or .5–.8;
- SEG: Number of segments, S = 2 or S = 5; and
- DIS: The distribution underlying the strength of the links, logistic and normal, which represent situations in which the distribution of the model is correctly specified versus misspecified.

We used a full 2^4 factorial design to generate data for nine countries, with 300 respondents per country, so that we had 2700 observations for each model run. In each cell of the design, we generated ten replications, which resulted in 160 data sets to be analyzed. The total number of links in the APT matrices was 18.

To generate the link probabilities, the u_{js}^m were taken from a fixed interval in such a way that the p_{js}^m were confined to the range dictated by the particular experimental condition. Then, according to the level of DIS, logistic or normal distributed errors were added to the u_{js}^m. The thresholds were drawn from a normal distribution, with standard errors $\sqrt{v_c}$ for each country, in the range dictated by the particular experimental condition. The μ_c were taken from an interval, such that the overall probabilities (given both u_{js}^m and μ_c) of choosing a link were in the interval (.05, .90). The prior segment proportions $\pi_{s|c}$ were taken from the interval (.15, .60). Then, the y_{ij}^m were generated as binary variables equaling 1 when $u_{js}^m > \theta_i$, and 0 otherwise (for all i and j, and i in segment s).

Our model is used to analyze each of the 160 synthetic data sets, using ten random starts to overcome local optima. As measures of model performance, we calculated the root mean squared error (RMSE) of the estimated versus the true parameters for p_{js}^m, $\pi_{s|c}$, μ_c, and $\sqrt{v_c}$, denoted by RMSE(p), RMSE(π), RMSE(μ), and RMSE($\sqrt{v}$), respectively. In addition, to assess how well the model identifies the segments from the data, we calculated the percentage of subjects that was assigned correctly to the segments (denoted by PRED). The optimal Bayes rule of classification is used, which assigns each subject to the segment with the largest posterior segment membership probability. The log or logit transformed RMSEs (and predictions) were subjected to analyses of variance to investigate the main effects and the interactions of PRB, TSE, SEG, and DIS.

For all measures, the F-ratios of the main and interaction effects, their significance, and R^2 values are reported in Table 1. We report the marginal means for each level of the factors in Table 2. The R^2 values in Table 1 show that, compared with the other measures of model performance, the variation of PRED is well explained by the experimental design (R^2 = .991), whereas the variation explained in RMSE(μ) is relatively low (R^2 = .273). The R^2 values for the other measures are close to .6.

Recovery of segment assignment (PRED) is significantly affected by the link probability interval (PRB), the number of segments (SEG), and the distribution of the strength of the links (DIS). The measure improves with increasing link probability intervals and decreases with the number of segments (Table 2), which is to be expected. However, the correct assignment in the S = 5 condition of 64.2% of the cases is still high, compared with a random assignment of 20%. Recovery of segment assignment decreases only slightly when the distribution is misspecified (a decrease from 77.8% to 76.4%). The PRB × SEG interaction in the ANOVA of PRED is the only sizable significant interaction in Table 1. Inspection of the marginal means in the PRB × SEG table indicates that the improvement of the predictions due to more variation in the link probabilities is much higher in the case of the S = 5 solution than in the case of the S = 2 solution. Apparently, when the number of segments increases, more distinctive patterns of the link probabilities improve classification of subjects to the segments more.

Analogous to PRED, recovery of the link probabilities, RMSE(p), and the segment proportions, RMSE(π), are significantly affected by PRB and SEG. They improve with larger intervals of the link probabilities and with fewer segments. Apparently, an increase in the probability interval and fewer segments helps the identification of the segments and, therefore, improves the recovery of these parameters. The magnitude of the threshold variance (TSE) and the distributional assumption (DIS) have no significant influence

on RMSE(p) and RMSE(π), which supports the robustness of the approach.

The recovery of the country-specific mean of the threshold parameters, RMSE(μ), is significantly affected by PRB and DIS. Its recovery improves with larger intervals of the link probabilities and correct distributional assumptions. The estimates of the standard errors of the thresholds, RMSE($\sqrt{v}$), are affected by PRB, TSE, and DIS. Larger intervals of link probabilities, larger variances of the threshold estimates, and correct underlying distribution of the link strengths have a positive significant effect on the recovery of the threshold variances. The precision of the threshold parameters, RMSE(μ) and RMSE($\sqrt{v}$), is not affected by the number of segments. This is caused by the number of threshold parameters being independent of the number of segments, as opposed to RMSE(p) and RMSE(π).

Most important, the overall means of the measures RMSE(p), RMSE(π), RMSE(μ), and RMSE($\sqrt{v}$) are low (.06, .05, .17, and .11, respectively), and the mean of PRED is high (77%). This means that the model recovers the true parameter values well and is effective in correctly assigning subjects to the segments. In general, we find that the structural part of the model is robust against misspecification of the distribution and response variance. As might be expected, the model performs better when the link probabilities have larger intervals, in particular for large numbers of segments. In our experience with APT in empirical studies, we typically find large variation in the link probabilities.

EMPIRICAL APPLICATION

Data Collection

Data were collected in an international survey in the EU as part of a major study on foods conducted for the European Commission. For the purpose of the application of our model, APT data on yogurt are used. The dairy industry is among the biggest food industries in the EU. On average, European consumers spend close to 15% of their overall food budget on dairy products, and annual sales of yogurt in the EU are approximately $10 billion. The yogurt market has been revitalized in the 1990s through new product development activity and heavy investments in branding and market communication, among other things. This has led to increasing differentiation of the market. The best market opportunities are expected from luxury/indulgence and enriched yogurts, whereas plain yogurt is declining. Yogurt is marketed by large multinational companies such as Nestlé, Danone, and Sodiaal-Yoplait, which operate at a European scale, as well as by smaller national companies. Companies also have set up joint ventures and strategic

Table 1
ANOVA RESULTS FOR MODEL PERFORMANCE MEASURES

Factor	*F-ratio PRED*[a]	*F-ratio RMSE(p)*[a]	*F-ratio RMSE(π)*[a]	*F-ratio RMSE(μ)*[b]	*F-ratio RMSE($\sqrt{v}$)*[b]
PRB	7556.345*	34.170*	103.330*	7.581*	51.054*
TSE	5.948	2.159	.393	1.225	33.920*
SEG	7295.315*	174.225*	98.294*	.054	2.364
DIS	42.476*	6.727	1.468	36.314*	87.026*
PRB × TSE	3.044	.299	1.716	.757	.270
PRB × SEG	884.132*	9.794*	.050	.061	.271
PRB × DIS	18.310*	.215	4.078	.810	.244
TSE × SEG	1.287	.049	.074	.057	.981
TSE × DIS	.415	.829	.598	.249	.107
SEG × DIS	7.097*	1.343	1.478	1.726	2.082
PRB × TSE × SEG	1.143	.328	.005	.251	.681
PRB × TSE × DIS	.472	1.574	.875	.030	.131
PRB × SEG × DIS	6.515	.655	3.090	.064	1.173
TSE × SEG × DIS	.003	.234	.081	.152	.902
PRB × TSE × SEG × DIS	1.233	.388	4.119	4.719	.165
R^2	.991*	.618*	.604*	.273*	.557*

[a]Transformed by logit.
[b]Transformed by log.
*$p < .01$.

Table 2
MEAN MODEL PERFORMANCE MEASURES ACCORDING TO DESIGN FACTORS

Factor	*Level*	*PRED(%)*	*RMSE(p)*	*RMSE(π)*	*RMSE(μ)*	*RMSE($\sqrt{v}$)*
PRB	[.3, .7]	63.9*	.070*	.070*	.186*	.131*
	[.1, .9]	90.4	.050	.040	.158	.098
TSE	[.3, .5]	77.3	.056	.054	.166	.129*
	[.5, .8]	76.9	.063	.056	.178	.100
SEG	S = 2	90.0*	.031*	.039*	.171	.111
	S = 5	64.2	.089	.071	.173	.118
DIS	logistic	77.8*	.057	.053	.140*	.093*
	normal	76.4	.062	.057	.204	.136

*Differences between means are significant on log or logit scale ($p < .01$).

Table 3
SAMPLE CHARACTERISTICS

	Response	*Sample Size*[a]	*Population Size*[b]	*Weights* w_c
Belgium	70%	272	4000	.343
Denmark	82%	229	2374	.242
France	82%	346	21,700	1.462
Germany	80%	261	32,200	2.876
Great Britain	69%	254	22,675	2.081
Greece	75%	292	3204	.256
Ireland	74%	211	1100	.122
Italy	59%	276	19,000	1.605
Netherlands	76%	260	6504	.583
Portugal	45%	267	3060	.267
Spain	69%	293	11,221	.893

[a]Number of households.
[b]Times 1000 households.

alliances to increase their market power in the yogurt market. For more details, see Valli, Traill, and Loader (1997).

For data collection, mail questionnaires were sent to households in 11 countries of the EU: Belgium, Denmark, France, Germany, Great Britain, Greece, Ireland, Italy, the Netherlands, Portugal, and Spain (i.e., all the EU-12 countries except for Luxembourg).[1] The fieldwork was carried out in 1996 by a pan-European marketing research agency. The questionnaires included the APT task for yogurt; consumption patterns; and sociodemographic, personality, and attitudinal variables, as well as additional measures that are not of interest to the purposes of this study.

Before collecting the pan-European data, extensive cross-national pretests were conducted. The relevant attributes and benefits were elicited through 100 in-depth laddering interviews (cf. Reynolds and Olson 1998) on consumers' MEC structures for yogurt. The resulting data were content analyzed into attributes, benefits, and values. These attributes, benefits, and values were included in APT matrices that were administered to consumers in Denmark, Greece, and Great Britain (N = 150). Based on frequency of occurrence, eight attributes and ten benefits were selected and included in the APT matrices that were used in the main study. The values used in APT are those from the "List of Values" (LOV) inventory (Kahle 1986). The LOV contains nine personal values that are relevant for consumer behavior (Kahle 1986; Kamakura and Novak 1992). These APT matrices were pretested among 48 Dutch consumers for wording, interpretability, and layout. Appropriate adjustments were made. Subsequently, an additional pretest was conducted in an international context using samples from France, Great Britain, Greece, the Netherlands, and Spain (N = 164). Again, appropriate adjustments were made and the final concepts included in the APT matrices were selected. Those concepts are shown in Appendix A. In the whole process, back-translation methods were used to ensure a similar content of the statements in the languages involved.

A sample, stratified by country, was drawn randomly from the household consumer panels of the pan-European market research agency. These panels are representative of the respective populations on a large number of sociodemographic characteristics. After sending reminders, the overall response was approximately 70%; 3147 questionnaires were obtained, from which 2961 were usable for analysis (see Table 3).

Results

We applied the segmentation model to the APT yogurt data. The model is estimated for S = 2 to S = 5. To overcome local optima, we ran each model from ten sets of random starting values. The solution with the maximum log-pseudo likelihood is reported. The information criteria for segment solutions S = 2 to S = 5 appear in Table 4. Both PBIC and PCAIC have minimal values for S = 4, which suggests that the four-segment solution provides the most parsimonious representation of the data. We denote the four segments S1, S2, S3, and S4. The entropy measures E_{Slc} are shown in Table 5. All values exceed .8, which indicates that in none of the countries have substantive segments been overlooked and that the segments are well separated.

In Table 5, we present the estimates of the country-specific segment proportions π_{slc}. The sizes of segments S1, S2, and S3 range from 15% to 20%. Segment S4 is by far the largest segment, comprising almost half the population. Segment S4 is represented in substantial proportion in all countries in the sample and is, therefore, a truly pan-European segment. Segments S1, S2, and S3 are cross-national segments, present in substantial proportions in multiple countries. These segments are not truly pan-European because they are not substantially represented in all countries. Segment S1 is mainly located along the coastal periphery of Europe (Denmark, Great Britain, Ireland, the Netherlands, Portugal, Greece, and Spain), consumers from Germany mainly dominate segment S2, and S3 is mainly located in the northwest of Europe (Great Britain, Germany, Denmark, Ireland, and the Netherlands). Segment S3, in particular, seems to demonstrate spatial correlation in being geographically contiguous.

The estimates of the response parameters are shown in the last two columns of Table 5. The mean thresholds, μ_c, vary over countries, being the highest in Great Britain and Denmark and the lowest in Italy and France. This means that, on average, in Great Britain and Denmark, respondents are least inclined to indicate links in the APT task, and in Italy and France, they are most inclined to do so. The within-

[1]The EU started in 1992 with 12 countries, which sometimes are denoted as the "EU-12" countries. Since then, Sweden, Austria, and Finland have joined the EU.

Table 4
SEGMENT SELECTION CRITERIA

S	*Log-Pseudo Likelihood*	*PCAIC*	*PBIC*
2	−167,908	338,657	338,341
3	−166,537	337,291	336,822
4	−164,950	335,494*	334,872*
5	−164,708	336,385	335,610

*Denotes minimum value.

country threshold variances, v_c, are similar in magnitude across countries. They have the smallest value in the Netherlands (.318) and the largest value in Denmark (.487); the Danish appear somewhat less homogeneous in their response tendency and the Dutch somewhat more. Compared with the absolute values of the mean thresholds, μ_c, the magnitude of the v_c is large, which indicates that within-country heterogeneity in response thresholds is larger than between-country heterogeneity.

The PCAIC and PBIC of the restricted models with equal threshold means across countries ($\mu_1 = \mu_2 = \ldots = \mu_C = 0$) are 335,584 and 334,972, respectively. For the model with no within-country threshold heterogeneity ($v_1 = v_2 = \ldots = v_C = 0$), these statistics are equal to 339,379 and 338,768, respectively. These information criteria are larger than those of the unrestricted model (335,494 and 334,872). This indicates that there are differences in response tendencies among, but also within, countries and supports the need for accommodating heterogeneity in excess of the level described by latent segments in standard mixture models.

Figures 1 through 4 show the MEC structures for the segments derived from the link strengths, represented graphically as "means–end maps." The attributes are located at the bottom of the figures (in arbitrary order), the benefits in the middle, and the values at the top. (Note that the concepts are abbreviated; see Appendix A for the complete formulations.) For each segment, lines represent the major links between attributes and benefits (AB links) and benefits and values (BV links). In addition, for each link, we present $p_{js}^m = \exp(u_{js}^m)/[1 + \exp(u_{js}^m)]$, the segment-specific probability that a link is chosen, adjusted for response tendencies (p_{js}^m is obtained from Equation 1, with $\theta_i = 0$ for all i and u_{js}^m substituted for u_j^m). To identify the major links, we tested whether each link's strength u_{js}^m is significantly larger than a chosen value θ^*. The lower the value of θ^*, the more the u_{js}^m will exceed the subject-specific thresholds. To provide a parsimonious representation and avoid clutter in the graphs, we chose $\theta^* = -.8$ as an appropriate cutoff level, corresponding to a minimal link probability of approximately .3. In the figures, the thickness of the lines reflects the corresponding link probability.

On examining the four MEC maps simultaneously, several APT links appear to be important across all four segments. Those links are important to all consumers in the 11 countries. Means–end chains that appear consistently across segments present opportunities for the development of standardized products that are supported by mass (unsegmented) communication. Of particular interest are complete chains, or unique perceptual orientations, of a particular attribute, benefit, and value, which present product positioning options. The chain "low fat–good for health–fun and enjoyment in life" is found in all four segments (though the connections are considerably stronger in S3 than in S1 and S4). This suggests opportunities for low-fat yogurt products that are mass-marketed to the EU consumers in general, with the resulting fun and enjoyment from good health as the consumer's driving value orientation in the communication campaign. The BV link between "good for your health" and "fun and enjoyment" is common across segments. However, major differences between the segments are also present.

Figure 1 displays the main links in the means–end structure for segment S1. Consumers in this segment do not have cognitive structures, regarding yogurt, as rich as those in the other segments. Seven AB links and four BV links appear in the map. Each attribute leads to one or two perceived benefits, but only one benefit connects to the value level. The central benefit is "good for your health," which is linked with four values. The attribute "low fat" leads to this central benefit. The four values present different positioning options by associating healthy, low-fat yogurt with security, hedonic, social, or self-respect values in advertising campaigns.

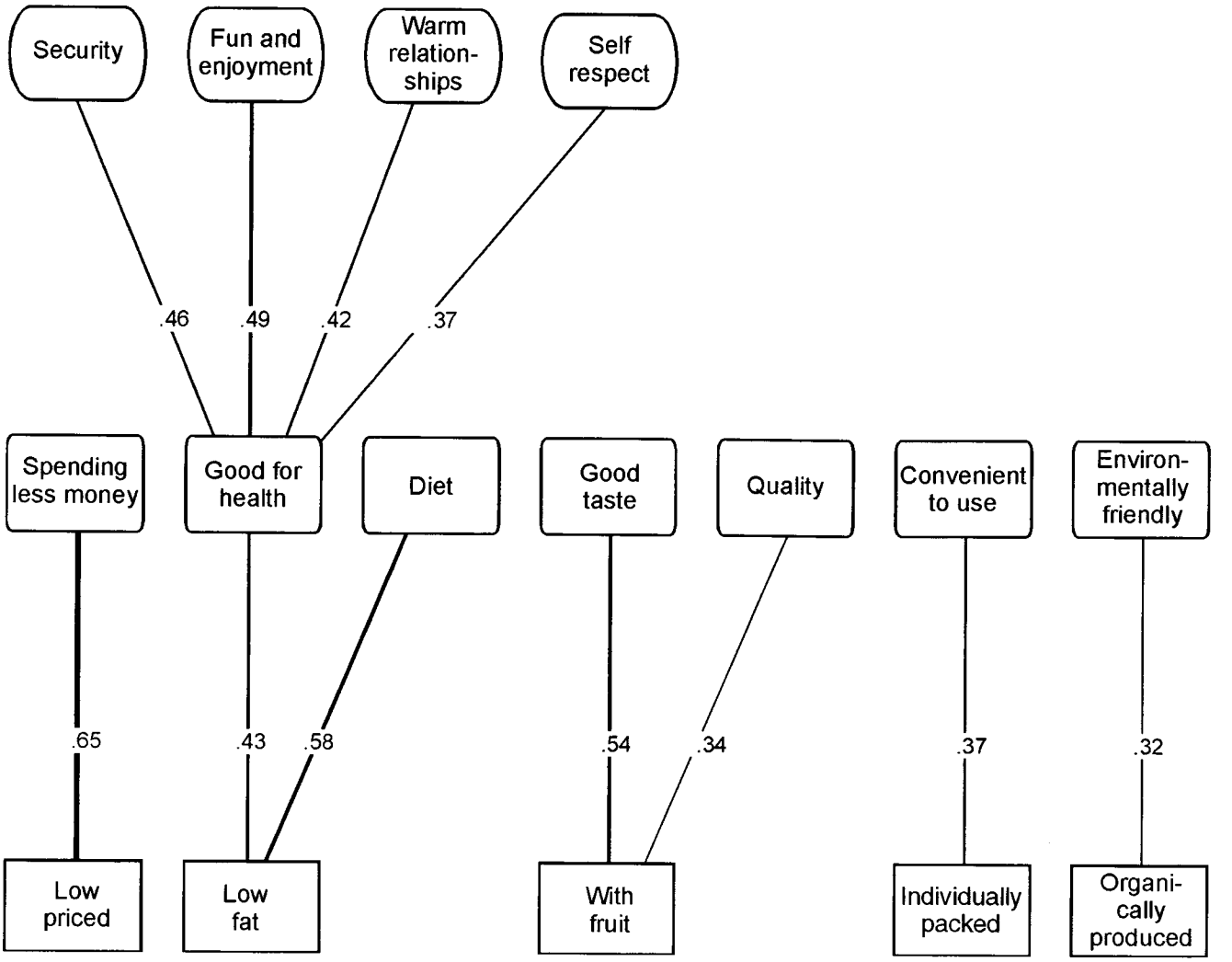
The MEC map of segment S2 in Figure 2 displays a rich structure of links: 16 AB links and 7 BV links arise. Multiple links connect to "good for your health" and "good quality," which seem to be central benefits for this segment. Subjects in this segment seem to categorize yogurt at a higher level of abstraction than do subjects in segment S1, in terms

Table 5
SEGMENT SIZES AND THRESHOLD ESTIMATES

	$E_{S\|c}$	$\hat{\pi}_{1\|c}$	$\hat{\pi}_{2\|c}$	$\hat{\pi}_{3\|c}$	$\hat{\pi}_{4\|c}$	$\hat{\mu}_c$	$\hat{v}_c$
Belgium	.876	12.2	17.5	8.5	61.8	−.017	.350
Denmark	.846	47.6	2.6	27.0	22.8	.173	.487
France	.899	2.4	21.8	3.3	72.4	−.145	.320
Germany	.826	3.2	45.1	26.3	25.4	−.018	.388
Great Britain	.807	41.2	7.0	26.0	25.9	.286	.339
Greece	.823	28.2	13.4	6.5	52.0	−.056	.466
Ireland	.806	40.6	12.1	17.9	29.5	−.045	.411
Italy	.896	5.9	10.2	5.1	78.8	−.228	.480
Netherlands	.821	31.5	14.3	17.9	36.3	.040	.318
Portugal	.835	35.2	27.8	4.9	32.1	−.009	.338
Spain	.832	25.1	8.5	3.8	62.6	.019	.361
Total*		17.4	20.9	15.8	46.0		

*The total is calculated as a weighted average of the country-specific priors.

Figure 1
PROBABILISTIC MEANS-END MAP SEGMENT S1

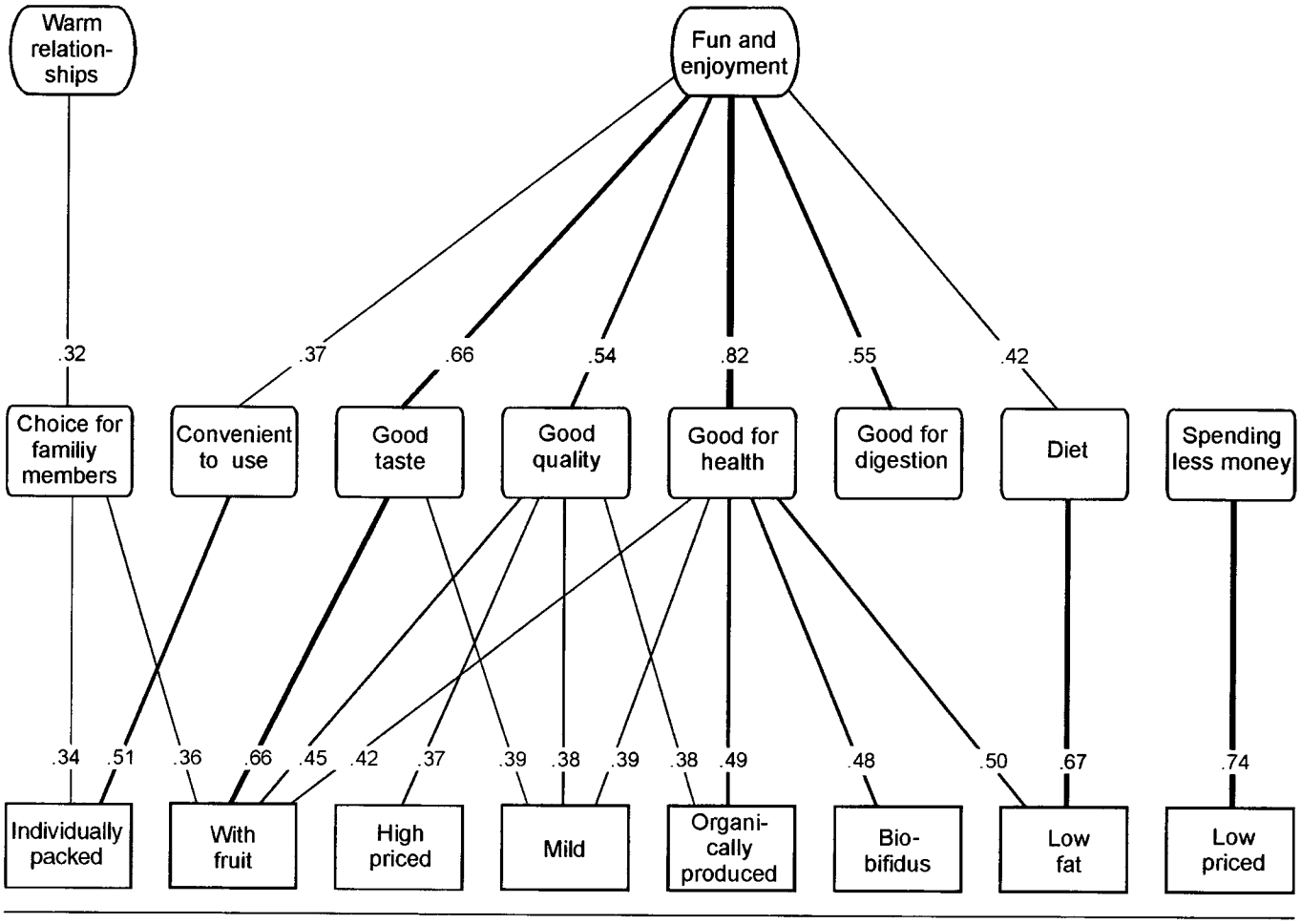


of the value level. "Fun and enjoyment in life" is clearly the need that this segment seeks to fulfill through the consumption of yogurt. Many benefits contribute to fulfillment of the need for fun and enjoyment, but especially "good taste" and "good for your health." A dominant MEC chain is the link between "with fruit," "good taste," and "fun and enjoyment in life" (both link probabilities are .66). In developing and communicating yogurt for this segment, fun and enjoyment in life should provide the major perceptual orientation. The multiple links with benefits, in particular with taste, quality, and health, and the large number of product attributes offering those benefits provide many opportunities for product development and positioning.

Consumers in segment S3 have many strong links, especially at the AB level. The MEC map in Figure 3 consists of 29 AB and 10 BV links. Consumers' cognitive structure with respect to yogurt appears to be more elaborate than in the two previous segments but, at the same time, seems to lack a single benefit or value that is central in the cognitive associative network. This suggests that multiple product options may be of interest to this segment. At the attribute level, "organically produced," "biobifidus," "low fat," and "with fruit" have many strongly associated benefits. In this segment, consumers derive health benefits from organically produced, low-fat yogurt and from yogurt with biobifidus. The health benefits "good for health," "good for digestion," and "replaces unhealthy snacks" have particularly rich associations with the attributes. Segment S3 shows the richest complete MECs involving diet. Low fat contributes to the product being "good if one is on a diet," which, in its turn, contributes to the three internally focused values (Homer and Kahle 1988) "self-respect," "sense of accomplishment," and "self-fulfillment." To compare, in S1 and S4, diet is not related to any value, whereas in S2, it is related to the interpersonal value (Homer and Kahle 1988) "fun and enjoyment."

Segment S4 contains fewer links than segments S2 and S3, but its cognitive structure is somewhat richer than that of segment S1. In the map, 12 AB links and 5 BV links are present (see Figure 4). Compared with the other segments, one of the distinguishing characteristics of this segment is the link between "good quality" and "security." Although the link probability is smaller than some of the other link proba-

Figure 2
PROBABILISTIC MEANS-END MAP SEGMENT S2



bilities, this link does not arise in any of the other segments. The attributes "high priced" and "organically produced" are linked to "good quality," forming a distinguishing MEC network that can serve as the basis for development of higher-priced, organic quality yogurts. In advertising campaigns, this product can be linked to security needs.

In summary, the results show distinct and actionable segments. In the cross-national segment S1, consumers typically are focused on the links between health benefits and various values, whereas the consumers in segment S2 use multiple benefits for the satisfaction of fun and enjoyment in life. The North-European segment, S3, typically derives a broad set of health-related benefits from multiple yogurt attributes, offering various (product development) means to the same benefit. The pan-European segment S4 displays much of the common MEC structure across segments, which is not surprising because it is the largest segment, but it has a distinguishing MEC component that centers around product quality.

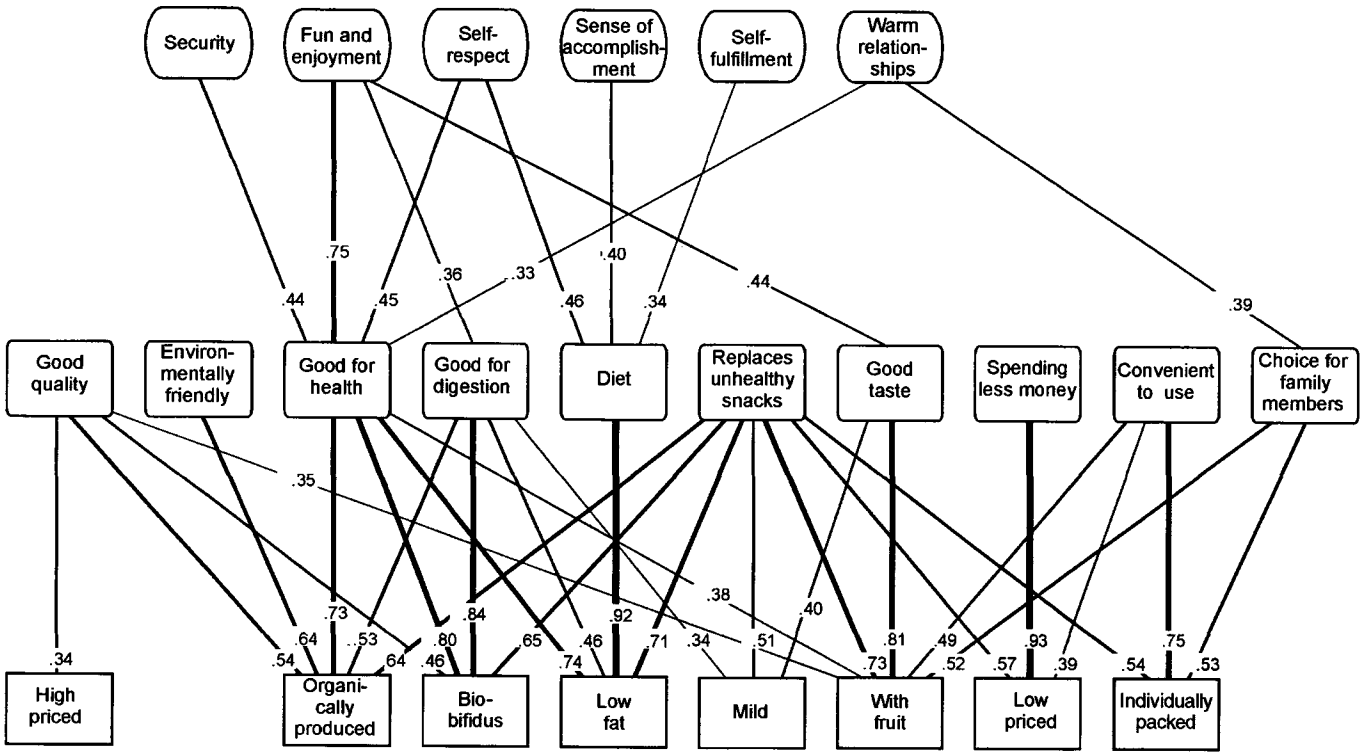
In the remainder of this section, we construct profiles of the segments by relating them to descriptive consumer data. Then, we assess the predictive validity and demonstrate the incremental contributions of our model over existing methods.

Segment Profiles

The segments were related to descriptor data, including sociodemographics, consumption patterns, information on media consumption, and information on personality and attitudes. The personality and attitudes constructs were deal proneness (cf. Lichtenstein, Netemeyer, and Burton 1995), consumer innovativeness (Steenkamp, ter Hofstede, and Wedel 1999), consumer ethnocentrism (Shimp and Sharma 1987), and product involvement (Mittal and Lee 1989). We constructed profiles of the four segments using Kamakura and Mazzon's (1991) regression-based procedure. The results, based on a 5% significant level, appear in Table 6.

Consumers in Segment S1 tend to be older and less educated, have lower incomes, and live in less urbanized areas. They spend less on yogurt and tend to visit smaller stores more often for their yogurt purchases. They tend to listen to the radio more often, watch more serials and entertainment programs, are more ethnocentric, and tend to be less innovative, deal prone, and involved with yogurt. The sociodemographic profile of S2 is similar to S1. Furthermore, these consumers use yogurt more often as a snack and tend to purchase it more often in larger stores. They tend to be more inclined to respond to promotions, less innovative, and more

Figure 3
PROBABILISTIC MEANS-END MAP SEGMENT S3



ethnocentric. Their attitudes and involvement regarding yogurt are generally higher. Segment S3 consists of younger consumers who are more educated and have higher incomes. They are relatively light users of yogurt but use it more often as a snack. These consumers are more exposed to radio and newspaper advertising. They are more deal prone and innovative, less ethnocentric, and have a more positive attitude toward yogurt in general. Consumers in Segment S4 have a sociodemographic profile similar to consumers in segment S3 but live more often in urban areas. They spend the most on yogurt, which they purchase more often in larger stores. Except for movies, these consumers are generally less exposed to media. In summary, the significant effects show a coherent pattern and contribute to the identifiability and accessibility of the segments.

Predictive Validity of the Model

To assess the predictive validity of our model, we cross-validated the results using holdout predictions of segment memberships. After assigning all respondents to the segment with largest posterior segment membership probability, estimation and holdout samples were constructed, and discriminant analysis was applied to the estimation samples. The discriminant function was constructed from the APT links to explain segment membership. The segment memberships in the holdout samples then were predicted using the discriminant function.

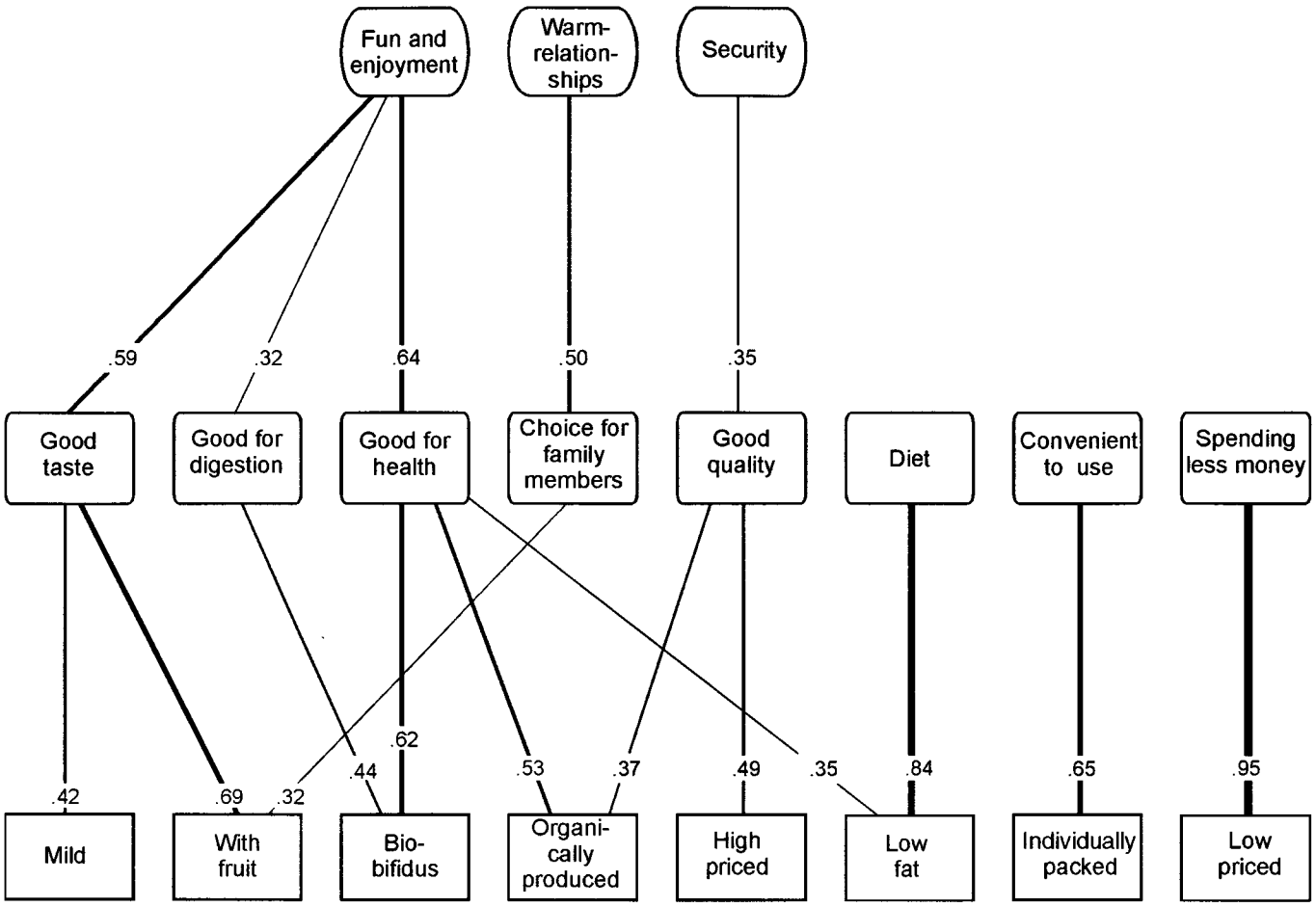
In predicting the segment memberships, two approaches have been taken. First, estimation and holdout samples were constructed by assigning, at random, two-thirds of the subjects to the estimation sample and one-third to the holdout sample. Second, we used the U-method (cf. Ghose 1998), which is related closely to the jackknifing method of classification. The U-method creates holdout samples of size one, so that holdout predictions are obtained for each individual separately. This procedure repeatedly omits one observation in estimation of the discriminant model and then predicts the segment membership of that omitted observation using the estimated discriminant function.

The procedures produced similar results. The first approach resulted in a prediction accuracy of 86%, ranging from 82% in S2 to 88% in S4. Thus, 87% of the cases are classified correctly, and no segment suffers from misclassification. The U-method resulted in a prediction accuracy of 87%, ranging from 82% in S2 to 91% in S4. The high prediction accuracy strongly supports the stability and predictive validity of the MEC segment structure identified with our model.

Comparison with Other Methods

Two alternatives for analyzing the pick-any data obtained from the APT tasks are correspondence and cluster analysis. Correspondence analysis relates to the nature of the binary APT data but does not arrive at the objective of our study, namely, identification of segments of consumers with similar MECs and construction of the MEC maps. Rather, it identifies continuous dimensions of subjects and concepts underlying the APT matrices. Cluster analysis, and in par-

Figure 4
PROBABILISTIC MEANS-END MAP SEGMENT S4



ticular K-means, is the prevailing approach to identify international segments (e.g., Kale 1995). One of the best clustering methods is K-means (Wedel and Kamakura 1998). To assess if, and to what extent, our model outperforms such a segmentation approach, we compare our model with K-means clustering.

Our model has several important features that are not accommodated by cluster analysis. First, a probabilistic classification method is used, as opposed to deterministic classification with K-means. Second, the model captures heterogeneity in response behavior, both within and between countries. Third, our model explicitly captures differences in segment sizes among countries, whereas K-means does not. Fourth, the pseudo likelihood is maximized, which accommodates the international sampling design and, therefore, leads to unbiased parameter estimates. K-means maximizes the ratio of the within-cluster and between-cluster variance and does not account for the sampling design. It may be expected that these features increase the performance of our model, compared with K-means.

We used the pan-European data set to compare our model with a K-means clustering solution. Three criteria for model comparison were used: model fit in terms of pseudo likelihood, correspondence of segment memberships, and the actionability of the segments. First, the fit of our model exceeds the fit of the K-means clustering solution for four segments. The difference in log-pseudo likelihood was a substantial 4691.12.

Second, the correspondence between the two solutions was low. We cross-classified the segments identified by the two methods.[2] The percentage of respondents that was classified by both approaches to the same segment (after appropriate permutation) ranged from approximately 40% for S1 to approximately 70% for S4 and was, on average, less than 50%. This is confirmed by the Rand index (RI; Rand 1971), RI = .597, which indicates that the probability that the methods allocate pairs of subjects in a similar way is approximately 60%. Note that this level of agreement is only slightly higher than that obtained when our method is compared with a random assignment of subjects (RI = .584).

Third, K-means clustering identified segments that were much less actionable. These segments discriminated mainly on the overall number of links that were chosen. We counted, for each segment, the number of times a link had the

[2]For our model, the subjects were classified according to the maximum a posteriori segment memberships.

highest probability compared with the other segments. For K-means, we arrive at 4%, 82%, 15%, and 0% of the links having maximal probability in clusters 1 to 4, respectively. For the solution of our model, these figures were 18%, 32%, 41%, and 9%. The more even distribution of links indicates that our model resulted in more actionable segments. Each segment had its own idiosyncratic pattern of links, whereas in the K-means solution, one segment (S2) in particular stands out because it provided an excessive number of dominant links. Apparently, these disappointing results of K-means are because K-means does not account for differences in response behavior and, therefore, leads to substantially different results.

DISCUSSION

In this study, we propose an approach to international market segmentation using MECs, which integrates international market segmentation and product positioning. Using MEC as a segmentation basis combines the advantage of product- and consumer-specific bases. It not only supports product positioning at the level of product attributes, but also includes the benefits derived by segments from those attributes, as well as the values satisfied. This increases the actionability of the identified segments for targeted strategies of international product development and communication. The proposed model provides segment-specific estimates of the strengths of links that tie the consumer to the product, that is, that identify cognitive associations between product attributes, benefits of product use, and consumer values at the segment level. It explicitly captures within- and between-country heterogeneity in response behavior. Heterogeneity is found to be highly significant both within and among countries.

We applied the methodology to identify segments in the European yogurt market, using a large cross-national sample of European consumers. Four segments were identified, of which one was truly pan-European. The segments were found to be actionable, and the pattern of links between attributes and values gave rise to strategic implications with respect to product development and communication. In addition, consumer characteristics were found to be related to segment membership, which contributes to the identifiability and accessibility of the segments. We also found high predictive accuracy in holdout samples, and an empirical comparison with K-means clustering demonstrated the better performance of our model.

Although our model was developed for segmentation in international markets, it can be adapted easily to the case of domestic segmentation, that is, segmentation within a single country. Our model nests a domestic segmentation model, which is obtained by imposing the following restrictions on the model: $\mu_c = 0$, $\nu_c = \nu$, and $\pi_{slc} = \pi_s$ for all $c = 1, ..., C$. These restrictions require dropping the mean threshold parameters in Equations 3 and 4 and concomitant variables in Equation 5 and estimating a single variance parameter for the thresholds in Equations 3 and 4. The log-pseudo likelihood in Equation 6 reduces to the standard log-likelihood, $\ln L_P = \Sigma_i \ln L_i$, and the PML approach reduces to a standard maximum likelihood approach.

Table 6
SEGMENT PROFILES

		Segment[b]			
Dependent Variable	R^2[a]	*S1*	*S2*	*S3*	*S4*
Sociodemographics					
Age	.114	+	+	−	−
Highest level of education	.148	−	−	+	+
Total household income after tax (in ECU)	.138	−	−	+	+
Size of the place of residence	.110	−			+
Consumption Patterns					
Expenditures on yogurt (in ECU)	.089	−		−	+
Frequency of using yogurt as a snack	.110	−	+	+	
Purchase frequency in convenience stores	.077	+			−
Purchase frequency in specialty stores	.077	+	−		
Purchase frequency in hypermarkets	.118	−	+	−	+
Media Consumption					
Frequency of listening to the radio	.105	+		+	−
Frequency of reading daily newspaper	.114		+	+	−
Frequency of watching serials	.152	+			−
Frequency of watching entertainment programs	.184	+			−
Frequency of watching movies	.084			−	+
Personality and Attitudes					
Deal proneness	.122	−	+	+	
Consumer innovativeness	.145		−	+	
Consumer ethnocentrism	.145	+	+	−	
Involvement with yogurt	.148	−	+		+
Overall attitude toward yogurt	.122	−	+	+	

[a]All R-values are significant at the .01 level.

[b]The plus and minus signs reflect the direction of the significant coefficients (at the 5% level) of regressing the dependent variables on the posterior segment memberships. The intercept was restricted to the mean value of the dependent variable. Therefore, a positive (negative) sign indicates that the behavior or attitude of consumers within a segment is greater (less) than average.

For international marketing purposes, the methodology lends itself to different types of international target market selection and differentiation strategies (which are not mutually exclusive; cf. Kotler 1997). First, one option is to develop specific products for specific segments. Second, a single segment may be targeted by a bundle of products (market specialization) to preempt competitive efforts and capture the variety-seeking tendencies of consumers (Baumgartner and Steenkamp 1996). Third, basically the same physical product can be developed for multiple segments. However, the product should be positioned differently in these segments, according to the segment-specific benefits and values to which the product's main attributes are perceived to lead. Such a product specialization strategy will result in substantial cost reductions in production. Fourth, a mass product, based on communalities in MECs among segments (if any), can be developed. Furthermore, our approach provides guidelines for integrated communication strategies by revealing meaningful linkages between MEC elements (Reynolds, Gengler, and Howard 1995).

Our MEC-based approach is useful not only for product positioning, but also as input for the new product development (NPD) process, which involves a range of decisions, from the identification of market opportunities to market forecast and communication. Our MEC-based approach is not the only input to this complex managerial process, which involves marketers, research and development personnel, production managers, and so on, but it may be an important one. A key aspect of NPD is the core benefit proposition, a short list of the strategic benefits that the new product provides to its customer segment and clues of how the product provides these benefits (Urban and Hauser 1993). Our procedure produces that information at the segment level. This information should be supplemented by, for example, cost and competitive information. Our methodology, based on MEC, is also a useful tool for coordinating thinking and communication across functional areas, which Griffin (1992) identifies as a major way to improve the NPD process.

Our analysis has been restricted to a single product category. Further research might focus on other categories and/or branded products to assess the pan-regional or global nature of the segments that our approach identifies. The inclusion of brands can be accommodated through an additional brand–attribute matrix in the sequential APT task (cf. Gutman 1982). However, it first must be assessed whether the brand perceptions in the brand–attributes matrix are independent of the links between attributes, benefits, and values. If so, the brand–attribute links may be added to the model (cf. ter Hofstede et al 1998). This is an issue for further research.

Another limitation of our empirical application is that our data were restricted to a limited set of countries, namely, 11 countries in the EU. Although there is significant cultural diversity among the European countries, the results cannot be generalized directly to other countries and do not provide an answer to the existence of global segments covering different parts of the world (e.g., Europe, Asia, Africa). Further research might focus on this issue and apply the methodology proposed in our study to countries from different continents.

We believe that the proposed methodology, which integrates international segmentation theory, measurement, and analysis in one framework, provides a step toward a closer fit of marketing research and managerial objectives in the international marketing domain.

APPENDIX A

In APT, respondents are confronted with two matrices, AB and BV. The matrices that have been used for the purpose of this study are given in Figure A1. In the AB matrix, a priori defined attributes and benefits compose the rows and columns, respectively. Similarly, the BV matrix includes benefits and values as rows and columns. For each column in the AB matrix (BV matrix), respondents indicate the benefits (values), if any, perceived to be associated with each of the product attributes (benefits).

The validity of APT to collect data on consumer MECs has been demonstrated by ter Hofstede and colleagues (1998). They show that the assumption of conditional independence of attributes and values, given benefits, holds across a wide range of conditions, which implies that an MEC relation can be measured in two steps through the AB and BV matrices. They also find that APT and the traditional qualitative laddering technique (cf. Reynolds and Olson 1998) display convergent validity, in that they yield the same MEC structure.

APPENDIX B

Following Bock and Lieberman (1970), the density unconditional on θ in Equation 3 may be approximated numerically by the following:

$$(B1)\qquad f_{ils}(y_i; u_s, \mu_c, v_c) = \left(\frac{1}{\sqrt{2\pi}}\right)\sum_{r=1}^{R} a_r \exp\left(\tfrac{1}{2} g_r^2\right) f_{ils}(y_i | g_r; u_s, \mu_c, v_c),$$

where a_r and g_r denote the abscissas and respective weights of the tables for the Rth order Gauss-Hermite quadrature. Substitution of Equation B1 in the log-pseudo likelihood in Equation 6 yields the following:

$$(B2)\qquad \ln L_P = \sum_{i=1}^{I} w_i \sum_{s=1}^{S} \pi_{slc(i)} \sum_{r=1}^{R} \tilde{a}_r f_{ils}\left(y_i | g_r; u_s, \mu_{c(i)}, v_{c(i)}\right),$$

where $\tilde{a}_r = a_r \exp(\tfrac{1}{2} g_r^2)$. The derivative of the log-pseudo likelihood with respect to γ_{sc} is equal to the following:

$$(B3)\qquad \frac{\partial \ln L_P}{\partial \gamma_{sc}} = \sum_{i:c(i)=c} w_i (\alpha_{is} - \pi_{slc}),$$

where α_{is} is the posterior probability obtained by Bayes rule. To ensure the threshold variances are positive, we use a reparameterization: $v_c = \exp(\tau_c)$. The derivatives of the log-pseudo likelihood with respect to u_{js}^m, μ_c, and τ_c are as follows:

Figure A1
ASSOCIATION PATTERN TECHNIQUE

AB MATRIX

	Choice for each member of the family	*Convenient to use*	*Environ-mentally friendly*	*Good for the digestion*	*Good for your health*	*Good if you are on a diet*	*Good quality*	*Good taste*	*Replaces unhealthy snacks*	*Spending less money*
High-priced yogurt	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Low-fat yogurt	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Low-priced yogurt	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Mild yogurt	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Organically produced yogurt	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Yogurt packed in individual portions	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Yogurt with biobifidus	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Yogurt with fruit	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

BV MATRIX

	Being well respected	*Excitement*	*Fun and enjoyment in life*	*Security*	*Self-fulfillment*	*Self-respect*	*Sense of accomplishment*	*Sense of belonging*	*Warm relationships with others*
Choice for each member of the family	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Convenient to use	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Environmentally friendly	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Good for the digestion	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Good for your health	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Good if you are on a diet	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Good quality	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Good taste	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Replaces unhealthy snacks	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Spending less money	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

$$(B4)\quad \frac{\partial L_P}{\partial u_{js}^m} = \sum_{i=1}^{I} w_i \frac{\pi_{s|c(i)}}{f_i\left(y_i; u, \mu_{c(i)}, v_{c(i)}\right)} \sum_{r=1}^{R} \tilde{a}_r f_{ir|s}\left(y_i|u_s, \mu_{c(i)}, v_{c(i)}\right) \left\{ y_{ij}^m - \frac{\exp\left[u_{js}^m - \exp(\tau_{c(i)})g_r - \mu_{c(i)}\right]}{1 + \exp\left[u_{js}^m - \exp(\tau_{c(i)})g_r - \mu_{c(i)}\right]} \right\};$$

$$(B5)\quad \frac{\partial L_P}{\partial \mu_c} = -\sum_{i:c(i)=c} \frac{w_i}{f_i(y_i; u, \mu_c, v_c)} \sum_{s=1}^{S} \pi_{s|c} \sum_{r=1}^{R} \tilde{a}_r f_{ir|s}(y_i|u_s, \mu_c, v_c) \sum_{m=1}^{2} \sum_{j=1}^{J_m} \left\{ y_{ij}^m - \frac{\exp\left[u_{js}^m - \exp(\tau_c)g_r - \mu_c\right]}{1 + \exp\left[u_{js}^m - \exp(\tau_c)g_r - \mu_c\right]} \right\};$$

and

$$(B6)\quad \frac{\partial L_P}{\partial \tau_c} = -\sum_{i:c(i)=c} w_i \frac{\exp(\tau_c)}{f_i(y_i; u, \mu_c, v_c)} \sum_{s=1}^{S} \pi_{s|c} \sum_{r=1}^{R} \tilde{a}_r f_{ir|s}(y_i|u_s, \mu_c, v_c) \sum_{m=1}^{2} \sum_{j=1}^{J_m} \left\{ y_{ij}^m - \frac{\exp\left[u_{js}^m - \exp(\tau_c)g_r - \mu_c\right]}{1 + \exp\left[u_{js}^m - \exp(\tau_c)g_r - \mu_c\right]} \right\}.$$

The PML estimates are obtained by equating Equations B4, B5, and B6 to 0 and solving for the unknown parameters.

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